



Project Overview

AirBnB is a global vacation rental platform that connects property owners (hosts) with travellers seeking accommodations. The platform features millions of listings, from budget-friendly apartments to luxurious villas. To remain competitive, AirBnB continuously analyses its data to improve user experiences, optimise pricing, and enhance host and guest satisfaction.

The analytics team at AirBnB is tasked with delivering insights into key performance areas, such as host activity, booking trends, and guest feedback. By leveraging the platform's data, they aim to optimise revenue, improve guest satisfaction, and identify areas of growth or underperformance.

Goal

The goal is to extract actionable insights from AirBnB's datasets, focusing on:

- ☐ Booking trends and occupancy rates.
- ☐ Host performance and listing popularity.
- ☐ Guest feedback and property diversity.

Dataset Description & Schema

AirBnB_Analysis/AirBnB Dataset.csv at main · shivangagrahari02-web/AirBnB_Analysis

Listings Table

Tracks detailed information about properties listed on AirBnB:

- **id**: Unique identifier for the listing.
- **name**: Title of the listing.
- **description**: Detailed property description.
- **host_id**: Identifier for the host managing the property.
- **host_name**: Name of the host.
- **property_type**: Type of property (e.g., house, apartment).
- **room_type**: Type of room (e.g., entire home, private room).
- **accommodates**: Maximum number of guests.
- **price**: Price per night.
- **review_scores_rating**: Average guest rating.
- **number_of_reviews**: Total number of reviews.

Calendar Table

Captures booking and availability data:

- **listing_id**: Foreign key linking to the id column in the listings table.
- **date**: Specific date for the entry.
- **available**: Indicates if the listing is available (t for true, f for false).
- **price**: Nightly price for the listing.

Reviews Table

Records guest feedback:

- **listing_id**: Foreign key linking to the id column in the listings table.
- **id**: Unique identifier for each review.
- **date**: Date the review was posted.
- **reviewer_id**: Identifier for the reviewer.
- **reviewer_name**: Name of the reviewer.
- **comments**: Text of the guest's feedback.

reviews		listings		calendar	
id 	int	id 	int	id 	int
listing_id	int	listing_url	text	listing_id	int
date	date	name	text	date	date
reviewer_id	int	description	text	available	boolean
reviewer_name	text	host_id	int	price	text
comments	text	host_name	text	minimum_nights	int
		host_since	date	maximum_nights	int
		host_location	text		
		host_response_time	text		
		host_response_rate	text		
		property_type	text		
		room_type	text		
		accommodates	int		
		bathrooms	decimal(3,1)		
		bathrooms_text	text		
		bedrooms	decimal(3,1)		
		beds	decimal(3,1)		
		price	text		
		number_of_reviews	int		
		review_scores_rating	decimal(3,2)		

Analysis & Queries

shivangagrahari02-web/ AirBnB_Analysis



AirBnB is a global rental platform that connects property owners (hosts) with travellers seeking accommodations. The platform features millions of...



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Contributor



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Issues



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Forks



Property Diversity

With a wide range of property types available, from apartments and villas to unique stays like treehouses and boats, the management team wants to evaluate whether they are meeting customer needs.

Highlighting the most common property types and uncover underserved categories.

1. The total number of listings and unique property types.



```
SELECT
```

```
  COUNT(id) AS total_number_of_listings,
```

```
  COUNT(DISTINCT property_type) AS unique_property_types
```

```
FROM listings;
```

	total_number_of_listing bigint	unique_property_type bigint
1	419	21

The platform offers a broad range of property types, indicating Airbnb's ability to serve multiple traveler segments ranging from budget travelers to luxury vacation seekers. However, if a small number of property types dominate inventory, Airbnb becomes vulnerable to shifts in traveler preferences. The supply acquisition team should target hosts operating in underrepresented categories such as cabins, cottages, or unique stays and aim to increase their inventory share by 10–15% over the next year to diversify platform offerings and reduce category concentration risk.

2. The top 5 most common property types and their respective counts.



```
SELECT
    property_type, COUNT(*) AS respective_counts
FROM listings
GROUP BY property_type
ORDER BY respective_counts DESC
LIMIT 5;
```

	property_type text	respective_counts bigint
1	Entire rental unit	203
2	Private room in home	67
3	Entire home	63
4	Private room in rental u...	19
5	Room in hotel	10

A significant portion of Airbnb inventory is concentrated within a few property categories, indicating where customer demand is strongest. These categories are likely responsible for a large share of bookings and revenue. The marketing team should prioritize these property types in paid acquisition campaigns while simultaneously launching host incentive programs for lower-supply categories. Success should be measured through growth in bookings for non-dominant property types and reduced dependence on the top categories.

Guest Ratings

Guest satisfaction is critical to Airbnb’s success, as high ratings lead to repeat bookings and positive word-of-mouth. Identifying top-rated listings can help Airbnb feature these properties in promotional campaigns and inspire other hosts to improve their offerings.

Conversely, understanding listings with poor ratings will allow Airbnb to address guest concerns through targeted interventions, such as host training or quality assurance checks.

1. Calculate the average review score across all listings.

	SELECT
	ROUND(AVG(review_scores_rating), 2) AS avg_review_score
	FROM listings;

	avg_review_score	
	numeric	
1	4.76	

The overall review score serves as a platform-wide customer satisfaction indicator. A high average rating suggests Airbnb is consistently meeting guest expectations, while a declining average would signal operational or service quality issues. The customer experience team should establish a minimum platform-wide rating benchmark and continuously monitor rating trends by geography and property type. A target improvement of at least 0.1 rating points annually can significantly improve booking conversion rates and guest trust.

2. List the top 10 listings with the highest review scores.

	SELECT
	name, review_scores_rating
	FROM listings
	WHERE review_scores_rating IS NOT NULL
	ORDER BY review_scores_rating DESC, name ASC
	LIMIT 10;

	name text	review_scores_rating numeric
1	Renovated 2br/2ba, close to Alb Med & St. Petes.	5
2	Quiet 2-bedroom condo in the heart of Albany	5
3	Luxury Albany Apt. Secure, Modern, w/ Parking	5
4	Spacious private suite and private bath	5
5	/zBig Blue Ranch\ 1962 SUNY Eagle Hill 4Beds 2B...	5
6	Elegant Studio in Center Square	5
Total rows: 10 Query complete 00:00:00.238		

Top-rated listings consistently deliver exceptional guest experiences and represent operational best practices. These listings can be used as benchmarks for host education programs. Airbnb should analyze common characteristics such as pricing strategy, amenities, response time, and property presentation, then incorporate these findings into host onboarding and training initiatives. Success can be measured by increasing the percentage of listings achieving ratings above 4.8.

3. Count the number of listings with an average review score below 4.0.



```
SELECT
    name,
    ROUND(AVG(review_scores_rating)::NUMERIC, 2) AS avg_rating
FROM listings
GROUP BY name
HAVING AVG(review_scores_rating) < 4.0;
```

	name text	round numeric
1	Central Albany Apartment: Walk to State Capitol!	3.96
2	Albany-3 bedrooms apt (2nd fl) A&A-Sisters Ren...	3.25
3	Private bedroom close to St. peter and Albany ...	3.67
4	Center Square Serenity: 2-Bedroom Elegance	3.50
5	Cozy 1 bedroom with office and luxurious King ...	3.00
6	/An Oasis 8 ppl Yard Grill W/D\	3.00
Total rows: 10 Query complete 00:00:00.144		

Low-rated listings represent a direct risk to guest satisfaction, repeat bookings, and platform reputation. These properties are likely experiencing issues related to cleanliness, communication, amenities, or expectation mismatch. Airbnb should create an automated intervention program that flags listings below 4.0 and requires corrective actions from hosts. Performance should be tracked by reducing the number of sub-4.0 listings by at least 20% within the next review cycle.

Host Engagement

Hosts are the backbone of Airbnb's platform, and their success directly impacts the company's growth. Hosts managing multiple listings contribute significantly to the platform's inventory and revenue. Airbnb wants to identify its most engaged hosts and understand their performance.

1. Identify hosts managing more than 3 listings.

	SELECT
	host_id, COUNT(id) AS number_of_listings
	FROM listings
	GROUP BY host_id
	HAVING COUNT(id) > 3
	ORDER BY number_of_listings DESC;

	host_id bigint 	number_of_listings bigint 
1	232967878	25
2	47625981	16
3	526358730	11
4	382970529	10
5	349460842	10
6	152762198	9

Hosts managing multiple listings contribute disproportionately to platform supply and likely represent professional operators rather than casual hosts. Their retention is strategically important because losing one such host can significantly reduce inventory. Airbnb should introduce a dedicated account management and support program for high-volume hosts and monitor host retention rates and listing growth within this segment.

2. Calculate the average review score for each host across their listings.

	SELECT
	host_id, id,
	ROUND(AVG(review_scores_rating)::NUMERIC, 2) AS avg_review_score
	FROM listings
	GROUP BY host_id, id
	ORDER BY avg_review_score;

	host_id bigint	id bigint	avg_review_score numeric
1	382446799	1190000000000000000	3.00
2	433884293	5590000000000000000	3.00
3	121680792	42389085	3.00
4	461579260	6870000000000000000	3.25
5	461579260	6820000000000000000	3.33
6	188679144	8430000000000000000	3.50
Total rows: 384		Query complete 00:00:00.115	

Host-level ratings provide a clearer measure of operational quality than individual listing ratings. Hosts consistently maintaining high ratings demonstrate scalable service quality across multiple properties. Airbnb should identify top-performing hosts and establish a mentorship framework where their practices are shared with newer hosts. Success should be measured through improvements in average host ratings among newly onboarded hosts.

3. List hosts with at least 2 listings and an average review score below 4.0.

	SELECT
	host_id, COUNT(host_id) AS total_listing,
	ROUND(AVG(review_scores_rating), 2) AS avg_rating_score
	FROM listings
	GROUP BY host_id
	HAVING COUNT(host_id) > 2 AND AVG(review_scores_rating) > 4.0
	ORDER BY avg_rating_score DESC;

	host_id bigint	total_listing bigint	avg_rating_score numeric
1	36438637	4	4.65
2	47625981	16	4.65
3	116935236	4	4.91
4	49100926	4	4.54
5	135717458	3	4.82
6	232967878	25	4.79
Total rows: 39		Query complete 00:00:00.107	

Hosts successfully managing multiple listings while maintaining strong guest ratings demonstrate an ability to scale operations without compromising customer experience. Airbnb should prioritize these hosts for Superhost recognition, promotional visibility, and beta-testing new host tools. Tracking booking growth and retention within this host segment will help measure the impact of these initiatives.

Booking Trends

Occupancy rates are one of the most critical indicators of a listing’s success. Listings with high occupancy rates indicate strong demand, while low-occupancy listings may signal pricing issues or uncompetitive offerings.

Airbnb’s sales team is keen to understand which listings perform well and which are underutilised during specific periods, such as January 2023. This insight will help them optimise marketing strategies and identify patterns in guest booking behaviour.

1. Calculate the occupancy rate for each listing for January 2023.

	SELECT
	id, host_since_date,
	((host_listings_count) / (host_total_listings_count)) * 100 AS occupancy_rate
	FROM listings
	WHERE host_since_date BETWEEN '2023-01-01' AND '2023-01-31';

	id bigint	host_since_date date	occupancy_rate integer
1	934000	2023-01-02	100
2	804000	2023-01-02	100
3	804000	2023-01-02	100
4	796000	2023-01-02	100
5	796000	2023-01-02	100
6	802000	2023-01-02	100
7	102000	2023-01-19	100
Total rows: 7		Query complete 00:00:00.121	

Occupancy rates reveal how effectively listings convert availability into bookings. High occupancy indicates strong demand, competitive pricing, and attractive property offerings, while low occupancy signals revenue leakage. Airbnb should identify common factors among high-performing listings and incorporate those factors into listing optimization recommendations. A measurable goal would be improving occupancy rates among bottom-quartile listings by 5–10%.

2. Identify the top 5 listings with the highest occupancy rates in January 2023.

	SELECT
	id, host_since_date,
	((host_listings_count) / (host_total_listings_count)) * 100 AS occupancy_rate
	FROM listings
	WHERE host_since_date BETWEEN '2023-01-01' AND '2023-01-31'
	ORDER BY occupancy_rate DESC
	LIMIT 5;

	id bigint	host_since_date date	occupancy_rate integer
1	934000	2023-01-02	100
2	804000	2023-01-02	100
3	804000	2023-01-02	100
4	796000	2023-01-02	100
5	796000	2023-01-02	100
Total rows: 5		Query complete 00:00:00.114	

The highest-occupancy listings demonstrate strong product-market fit and effective pricing strategies. Understanding what differentiates these properties can help Airbnb improve performance across the broader marketplace. The host success team should conduct comparative analysis of amenities, pricing, and location attributes and use the findings to create optimization guidelines for underperforming listings.

3. List all listings that were not booked at all in January 2023.

	SELECT
	listing_id
	FROM calender
	WHERE date2 BETWEEN '2024-01-01' AND '2024-01-31'
	GROUP BY listing_id
	HAVING SUM(CASE WHEN available = 't' THEN 1 ELSE 0 END) = 0;

Listings with no bookings represent unrealized inventory and lost revenue opportunities. These listings may suffer from poor visibility, weak reviews, uncompetitive pricing, or incomplete profiles. Airbnb

should launch an automated diagnostic program that identifies the primary reason for underperformance and recommends corrective actions. Success should be measured through reductions in inactive listings and improvements in first-booking conversion rates.

Pricing Patterns Across Property Types

Airbnb's pricing strategy team wants to understand how property type influences nightly rates. Identifying undervalued property types can guide pricing adjustments, while understanding high-value categories can inform marketing priorities.

Analysing pricing patterns will also help identify trends in demand and inform strategic decisions about property offerings.

1. Calculate the average price per night for each property type.

	SELECT
	listing_id,
	ROUND(AVG(price)::NUMERIC, 2) AS average_price_per_night
	FROM calender
	GROUP BY listing_id;

	listing_id bigint	average_price_per_night numeric 
1	437362	179.00
2	881000	89.00
3	662333	125.00
4	807000	100.00
5	849000	270.00

Average nightly pricing provides insight into market positioning and affordability. Significant pricing variation suggests the existence of multiple customer segments with different willingness to pay. Airbnb should use these findings to strengthen its dynamic pricing engine and improve price recommendations. Performance should be measured through increases in revenue per available night while maintaining occupancy levels.

2. Identify the top 5 listings with the highest average price per night and their property type.



```
SELECT

C.listing_id,

ROUND(AVG(C.price)::NUMERIC, 2) AS avg_price_per_night,

L.property_type

FROM calender AS C

JOIN listings AS L

ON C.listing_id = L.id

GROUP BY C.listing_id, L.property_type

ORDER BY avg_price_per_night DESC LIMIT 5;
```

	listing_id bigint	avg_price_per_night numeric	property_type text
1	929000	4000.00	Entire rental unit
2	622000	1350.00	Entire home
3	117000	874.50	Entire rental unit
4	117000	874.50	Entire home
5	117000	874.50	Private room in ho...
Total rows: 5		Query complete 00:00:00.444	

Premium-priced listings target high-value travelers and contribute disproportionately to revenue despite lower booking volumes. These properties require exceptional guest experiences to justify their pricing. Airbnb should develop luxury-focused marketing campaigns and monitor booking conversion rates within the premium segment to ensure pricing remains aligned with customer expectations.

3. Find property types with an average price below \$150 per night.

	SELECT
	L.property_type,
	ROUND(AVG(C.price)::NUMERIC, 2) AS avg_price_per_night
	FROM calender AS C
	JOIN listings AS L
	ON C.listing_id = L.id
	GROUP BY L.property_type
	HAVING AVG(C.price) < 150.0;

	property_type text	avg_price_per_night numeric
1	Entire condo	120.86
2	Entire guest suite	91.00
3	Entire loft	115.33
4	Entire serviced apartm...	118.50
5	Entire townhouse	131.92
Total rows: 13		Query complete 00:00:00.269

Property types averaging below \$150 per night appeal primarily to budget-conscious travelers and represent an important source of booking volume. Airbnb should highlight these categories in affordability-focused campaigns and introduce dedicated budget traveler filters. Success should be measured through increased booking volumes and improved occupancy within these property segments.

Guest Review Insights

Guest reviews offer invaluable insights into the strengths and weaknesses of Airbnb's platform. Identifying frequent reviewers can help the company understand customer behaviour, while analysing listings with no reviews reveals areas of potential improvement in guest engagement.

1. Identify the top 10 reviewers by the total number of reviews submitted.

	SELECT
	reviewer_id,
	COUNT(*) AS total_reviews
	FROM reviews
	GROUP BY reviewer_id
	ORDER BY total_reviews DESC
	LIMIT 10;

	reviewer_id  bigint	count  bigint
1	414682721	93
2	129172443	53
3	2450995	29
4	17224232	29
5	5373770	27

Frequent reviewers are highly engaged customers who provide valuable feedback and contribute significantly to platform trust. Airbnb should establish a reviewer recognition program that rewards consistent review participation. Success should be measured through increases in review submission rates and overall review coverage across completed stays.

2. Calculate the average number of reviews per listing.

	SELECT
	id,
	ROUND(AVG(number_of_reviews), 2) AS avg_no_of_reviews
	FROM listings
	GROUP BY id;

	id bigint	avg_no_of_reviews numeric
1	437362	123.00
2	881000	11.00
3	662333	328.00
4	807000	32.00
5	849000	1.00

Listings with higher review volumes generally enjoy stronger customer trust and better booking conversion. Airbnb should focus on increasing review generation through automated reminders and post-stay engagement strategies. A measurable target would be increasing average reviews per listing by 15% annually.

3. Identify all listings with no reviews in 2023.

	SELECT
	id
	FROM listings
	WHERE host_since_date BETWEEN '2023-01-01' AND '2023-12-31'
	GROUP BY id
	HAVING SUM(number_of_reviews) = 0;

Listings without reviews face a credibility disadvantage because potential guests lack social proof. Airbnb should prioritize these listings in review-generation campaigns and encourage hosts to provide exceptional first-stay experiences. Success should be measured by reducing the percentage of review-less listings and increasing first-review acquisition rates.

Customer Engagement Trends

The team wants to analyze customer engagement across all listings within a specific time period. By identifying and ranking reviews submitted between two dates, the business can understand trends in customer interactions and review activity during that timeframe.

Rank all reviews submitted between '2015-01-01' and '2015-06-30' by their submission date.



```
SELECT
    listing_id,
    date2 AS review_date,
    reviewer_name,
    DENSE_RANK() OVER ( ORDER BY date2) AS rank_of_review_by_date
FROM reviews
WHERE date2 BETWEEN '2015-01-01' AND '2015-06-30';
```

	listing_id bigint	review_date date	reviewer_name text	rank_of_review_by_date bigint
1	382021	2015-01-01	Molly	1
2	299245	2015-03-04	Altay	2
3	299245	2015-03-25	John	3
4	382021	2015-04-16	Macy	4
5	382021	2015-04-18	Ben	5
Total rows: 37		Query complete 00:00:00.110		

Review activity trends provide insight into periods of peak customer engagement and booking activity. Higher review volumes during specific periods indicate seasonal demand patterns and marketing effectiveness. Airbnb should align promotional campaigns with historically high-engagement periods and monitor review growth as a leading indicator of marketplace activity.

Most Active Hosts

The management team wants to identify the most active hosts based on the total number of reviews their listings have received. This analysis will help the team recognize high-performing hosts who maintain popular and engaging properties. The total number of reviews for each host is calculated by summing up the reviews of all their listings.

Calculate the total number of reviews received by each host across all their listings. Rank the hosts based on the total number of reviews.



```
SELECT

    host_id,

    host_name,

    SUM(number_of_reviews) AS total_number_of_reviews,

    DENSE_RANK() OVER ( ORDER BY SUM(number_of_reviews) DESC ) AS rank_of_host

FROM listings

GROUP BY host_id, id, host_name;
```

	host_id bigint	host_name text	total_number_of_reviews bigint	rank_of_host bigint
1	9666970	Janet	853	1
2	5691268	Elizabeth	697	2
3	44434571	Earl	648	3
4	17766924	Sugey	598	4
5	44434571	Earl	574	5
Total rows: 384		Query complete 00:00:00.172		

Hosts receiving the highest volume of reviews are likely generating the most guest interactions and maintaining strong booking activity. These hosts are critical contributors to platform engagement and revenue generation. Airbnb should focus retention efforts on this segment and track host churn rates, review growth, and booking performance to ensure continued marketplace stability.

Revenue for Top Listings

The finance team wants to identify the top revenue-generating listings. By analyzing the total revenue earned by each listing, the team can prioritize these listings for promotional strategies or investment. The revenue is calculated by summing up the price for all available days in the calendars table.

Calculate the total revenue for each listing by summing up the price for all available days. Use a window function to rank the listings based on their total revenue in descending order.

	SELECT
	listing_id,
	SUM((price) * (CASE WHEN available = 't' THEN 1 else 0 end)) AS Total_revenue,
	DENSE_RANK () OVER (ORDER BY SUM((price) * (CASE WHEN available = 't' THEN 1 ELSE 0
	END)) DESC) AS Rank_of_list_by_revenue
	FROM calender
	GROUP BY listing_id

	listing_id bigint	total_revenue numeric	rank_of_list_by_revenue bigint
1	117000	1109036.00	1
2	929000	840000.00	2
3	938000	575546.00	3
4	622000	448200.00	4
5	101000	324861.00	5
Total rows: 301		Query complete 00:00:00.136	

Revenue concentration among a small number of listings indicates that specific property characteristics, locations, or host practices significantly outperform the marketplace average. Airbnb should identify the key drivers behind these listings and build data-driven recommendations for other hosts. Success should be measured through revenue growth among mid-tier listings adopting similar practices.

Property Type Performance in Summer

The marketing team wants to identify how property types perform relative to each other during the summer months. By calculating percentile ranks for property types based on their total summer revenue, the team can classify property types into performance tiers, such as top-80 percentile performing property types.

Calculate the total summer revenue (June, July, and August) for each property type. Determine its percentile rank among all property types based on their revenue.



```
SELECT

    L.property_type,

    SUM((C.price)*(CASE WHEN C.available ='t' THEN 1 ELSE 0 END)) AS Total_revenue,

    PERCENT_RANK () OVER(ORDER BY SUM((C.price)*(CASE WHEN C.available ='t' THEN 1 ELSE
0 END ))) *100 AS Percentile_Rank

FROM listings AS L

JOIN calender AS C

ON L.id = C.listing_id

WHERE EXTRACT(MONTH FROM date2) IN (6, 7, 8)

GROUP BY L.property_type
```

	listing_id bigint	total_revenue numeric 🔒	rank_of_list_by_revenue bigint 🔒
1	117000	1109036.00	1
2	929000	840000.00	2
3	938000	575546.00	3
4	622000	448200.00	4
5	101000	324861.00	5
Total rows: 301		Query complete 00:00:00.191	

Certain property types generate substantially higher revenue during summer months, reflecting strong seasonal demand patterns. Airbnb should increase promotional investments for these property categories before peak travel periods and encourage hosts to optimize pricing during high-demand seasons. Performance should be measured through seasonal revenue growth and booking share gains within targeted property types.

Year-over-Year (YoY) Growth in Listing Revenue

The finance team wants to analyze how listing revenue changes over time by comparing year-over-year (YoY) growth. This information can help identify trends in revenue generation and highlight listings that are growing or declining in performance.

For each listing, calculate the total revenue per year and determine its YoY growth as a percentage. Use window functions to retrieve the previous year's revenue for each listing and calculate the YoY growth. Finally, rank listings based on their YoY growth percentage to identify top-performing listings.



```
WITH my_cte AS (  
  
    SELECT  
  
        listing_id,  
  
        EXTRACT(Year FROM date2) AS year,  
  
        SUM((price) * (CASE WHEN available = 't' THEN 1 ELSE 0 END)) AS Total_revenue  
  
    FROM calender  
  
    GROUP BY year, listing_id  
  
    ORDER BY listing_id, year)  
  
SELECT *,  
  
    LAG(total_revenue,1) OVER(PARTITION BY listing_id) AS pre_value,  
  
    ROUND((Total_revenue - (LAG(total_revenue,1) OVER(PARTITION BY  
listing_id))) * 100 / NULLIF(LAG(total_revenue,1) OVER(PARTITION BY listing_id), 0), 2) AS YOY_growth  
  
FROM my_cte
```

	listing_id bigint	year numeric	total_revenue numeric	pre_value numeric	yoy_growth numeric
1	299245	2023	0.00	[null]	[null]
2	299245	2024	11060.00	0.00	[null]
3	382021	2023	11000.00	[null]	[null]
4	382021	2024	31000.00	11000.00	181.82
5	565157	2023	2550.00	[null]	[null]
Total rows: 602		Query complete 00:00:00.230			

Listings demonstrating strong YoY revenue growth indicate successful pricing, marketing, or operational strategies, while declining listings signal potential competitive or quality concerns. Airbnb

should use high-growth listings as benchmark cases and proactively investigate declining properties. Success should be measured by increasing the proportion of listings achieving positive YoY revenue growth and reducing the number experiencing revenue decline.